

ARTICLE

A Ground–Low-Altitude Mobility Manager for Service Reconfiguration under Disruptions

Author Name^a

^aDepartment, University, City, Country

ARTICLE HISTORY

Compiled September 13, 2026

ABSTRACT

Low-altitude capacity alone does not ensure service resilience: an intervention must preserve useful service before its deadline margin is exhausted. We propose a Ground–Low-Altitude Mobility Manager that organizes joint ground–air states into a shared executable candidate interface. Replaceable supervisory policies select interventions using common feasibility, completion-time and incumbent-service facts, while execution checks and independent local contingencies govern their application. Four SUMO–BlueSky experiments across three contrasting transport settings examine selective air support, air failures, compound disturbances and observation–execution timing. At the meshed urban site, the fixed-objective heuristic and LLM-candidate policy both reduce emergency deadline violations from 33.33% to 16.67%, but leave 0.058 and 0.133 damaged incumbent services per run, respectively. Explicit candidate-information support changes choices when faster air service competes with service preservation. Every affected coordinated run restores a path after a single air failure, yet 39.06% of all LLM-policy runs at that site remain late. Delayed observation consumes the remaining margin, and execution waiting can change the intervention itself through command replacement. The study establishes an executable architecture for comparative service management and identifies surviving alternatives, incumbent-service costs and intervention timing as the mechanisms governing realized resilience. The management implication is to secure timely, useful reconfiguration, rather than equating available air capacity or a restored path with recovered service.

KEYWORDS

ground–low-altitude mobility; service reconfiguration; disruption management; supervisory control; transport resilience; network morphology.

1. Introduction

Restoring network connectivity does not necessarily restore transport service. An urgent mission may retain a road route but lose enough time to miss its deadline. A useful response must therefore preserve completion within the remaining service margin. Reassigning a vehicle can rescue that mission while interrupting another service. This links disruption management to both deadline compliance and the distribution of service losses, consistent with performance-based transport resilience assessment. (Mattsson and Jenelius 2015)

Low-altitude resources offer a way around disrupted ground links. Their value depends

on where they are, which tasks they can serve and whether they are already occupied. A faster aircraft may require preempting an incumbent service even when continued ground transport would remain timely. Communication loss, degraded navigation and unavailable landing sites can also remove the air alternative after dispatch. The relevant management problem is thus to select and execute useful cross-modal interventions as resources and deadlines evolve.

Multimodal recovery and Resilience as a Service already coordinate heterogeneous replacement services, while dynamic fleet management and truck–drone routing address evolving assignments and synchronization. (Pillac et al. 2013; Yap et al. 2022; Jaber et al. 2025; Meng et al. 2024; Wollenstein-Betech et al. 2022) Constrained-execution architectures further distinguish a proposed action from permission to apply it. (Wabersich and Zeilinger 2021) We connect these ideas at the level of supervisory service reconfiguration: a failure-prone air layer supports disrupted ground services through a common interface that exposes feasible interventions and their consequences. This creates a controlled basis for comparing selection policies and tracing why an admissible replacement sometimes arrives too late.

We propose the **Ground–Low-Altitude Mobility Manager** to connect joint state observations, deterministic candidate generation, replaceable supervisory selection and common execution checks. Its **shared executable candidate interface** reports resource compatibility, legality, estimated completion and the incumbent service displaced by each option. The policy selects one intervention; the executor normalizes and validates it, then issues it or queues it for revalidation. Local contingencies operate independently. Here, global state denotes the joint state of one modeled study area. Ground service uses a SUMO corridor ETA timer; air transport evolves in BlueSky.

We make three contributions. First, we develop a ground–air supervisory service-reconfiguration architecture with explicit observation, selection and execution semantics. Second, we provide a shared executable candidate interface and comparative evidence from ground-only, air-first, fixed-objective heuristic and LLM-candidate policies. Third, we quantify transport mechanisms linking surviving alternatives, service trade-offs and observation–execution timing. Selective air support reduces missed deadlines but can displace incumbent services; restored paths can remain late; delayed execution can replace the originally selected action. Figure 1 connects these findings to E1 selective support, E2 single air failure, E3 compound disturbance and E4 timing and input burden.

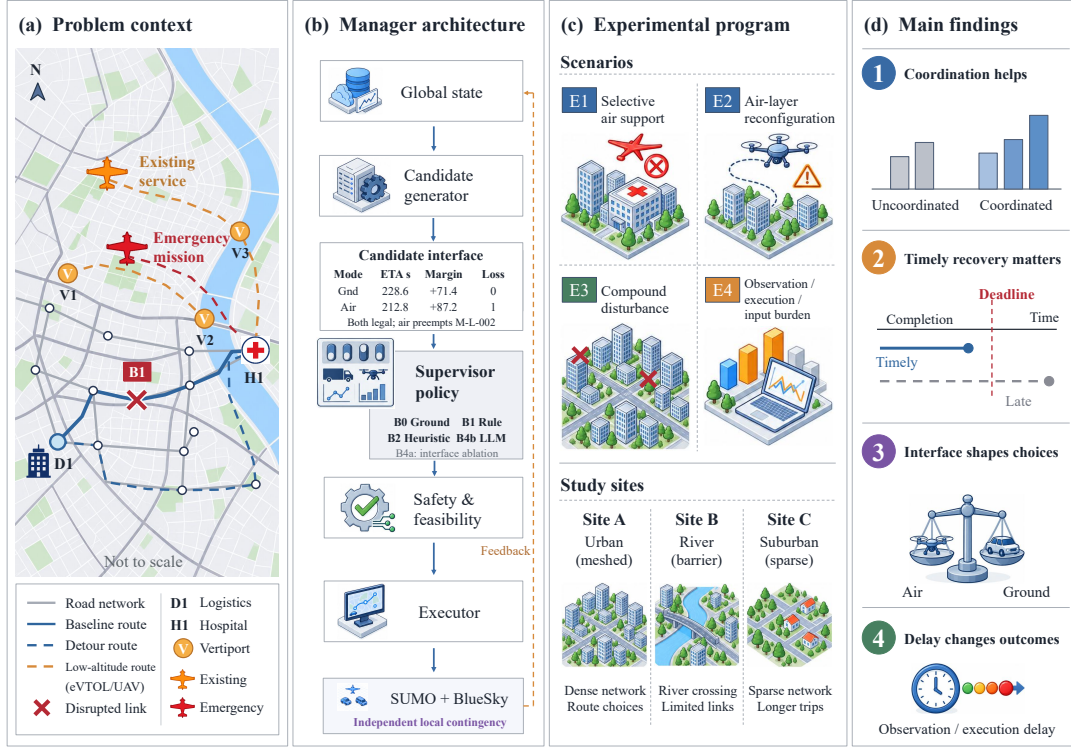


Figure 1. Overview of the Ground-Low-Altitude Mobility Manager. (a) Schematic service network with logistics D1, hospital H1, landing nodes V1–V3, ground and low-altitude routes, and a disrupted link B1. The schematic map is not to scale; map B1 is distinct from policy B1. (b) Shared candidate facts connect replaceable supervisory policies to modeled feasibility checks and execution. B0/B1/B2/B4b are primary policies; B4a is an E1 interface ablation. (c) E1 examines selective air support, E2 air-layer reconfiguration, E3 compound disturbance and E4 observation, execution and input burden. E1–E3 use sites A–C; E4 uses A. (d) Conceptual findings express conditional coordination benefit, timely recovery, interface-dependent choices and delay effects. The interface effect is strongest at A. Illustrations are schematic; symbols and bars are qualitative, not measured outcomes.

2. Related work

2.1. Disruption management across transport services

Transport vulnerability and resilience research connects component disruptions to changes in system performance. Mattsson and Jenelius distinguish the related concepts, while Jenelius and Mattsson explain how network vulnerability measures can be implemented. (Mattsson and Jenelius 2015), (Jenelius and Mattsson 2015) Cats and Jenelius show why partial capacity degradation also matters for public transport performance. (Cats and Jenelius 2018) These studies motivate evaluating service completion and deadline compliance alongside the existence of an alternative route.

Operational responses can propagate between transport services. Lai and Leung study real-time public transit rescheduling under disruption. (Lai and Leung 2018) Yap et al. couple train rescheduling with dynamic passenger assignment to represent disruption propagation into urban tram and bus networks. (Yap et al. 2022) Their framework makes effects outside the initially disrupted layer part of the response assessment. Here, incumbent-service damage provides a narrower service-level account of such consequences, without modeling passenger redistribution across a public transport network.

Jaber et al. formulate Resilience as a Service through resource allocation among buses, taxis and automated vans that can bridge disrupted rail services. (Jaber et al. 2025) Their work establishes a close precedent for coordinating heterogeneous replacement services. Our setting adds a failure-prone air layer and tests replaceable supervisory policies through a shared executable candidate interface. The common interface allows this failure-prone support layer to be studied through interchangeable selection policies and their executed service consequences.

2.2. *Dynamic fleet management and ground–air coordination*

Dynamic vehicle routing addresses assignment and routing decisions as operational information changes. (Pillac et al. 2013) Ground–air coordination introduces resource synchronization: the flying sidekick traveling salesman problem jointly plans truck and drone movements for parcel delivery. (Murray and Chu 2015) Meng et al. incorporate stochastic truck travel times and soft time windows, using rolling-horizon recourse to update truck waiting times during execution. (Meng et al. 2024) Their work illustrates that timing and synchronization are substantive transport decisions, beyond selecting the nominally fastest mode.

At the network level, Wollenstein-Betech et al. jointly optimize intermodal vehicle routing and rebalancing while accounting for congestion generated by vehicle flows. (Wollenstein-Betech et al. 2022) The present model uses a smaller decision object: an intervention for an actionable service, with ground supplied through a shared corridor travel-time proxy. It does not optimize network-wide routing flows or truck–drone launch-and-retrieval tours. Its comparison concerns how policies select, preserve or replace services as modeled feasibility changes. B2 operationalizes one explicit speed–service-preservation preference through one-step candidate ranking.

2.3. *Supervisory selection and constrained execution*

Urban air mobility concepts distinguish vehicle operations, operating locations and supporting management services. (Federal Aviation Administration 2023) More generally, control architectures can separate a proposed action from the mechanism permitting execution. Wabersich and Zeilinger’s predictive safety filter exemplifies this separation under a specific constrained-control formulation. (Wabersich and Zeilinger 2021) We adopt the proposal–execution separation and implement transport-specific deterministic rule checks.

Our candidate generator exposes common feasibility, completion-time and incumbent-service facts before selection. The executor subsequently checks the chosen action against its current state. This distinction supports a transport comparison in which policy preferences can differ while information derivation and execution constraints remain common. It also exposes the interval between an admissible proposal and the intervention that actually reaches execution.

2.4. *Language models and the comparison addressed here*

SayCan connects language-based choices to executable robotic capabilities, while LLM-Light applies language-based decisions to traffic signal control. (Ichter et al. 2023; Lai et al. 2024) These studies motivate an LLM as one implementation of the supervisory policy. Long-context research also distinguishes information availability from effective

use. (Liu et al. 2024)

Our comparison asks how policies use a common service-management contract and when their executed interventions protect service. B4a–B4b separately tests explicit candidate-information support within the same LLM implementation. It removes derived per-candidate facts along with their presentation, so its effect concerns information support rather than table formatting alone.

3. Problem formulation

3.1. *Ground–low-altitude service system*

Let the ground network be a directed graph $G^g = (V^g, E^g)$. The low-altitude layer contains aircraft, operating sites, assignments, routes and active restrictions. Time advances in discrete simulation steps. At time t , the combined service state is

$$S_t = (G_t, A_t, I_t, M_t, E_t), \quad (1)$$

where G_t contains ground connectivity and travel-time estimates, A_t contains aircraft states, I_t describes landing infrastructure and operating restrictions, M_t is the mission registry, and E_t records currently represented disturbances and recent events. A mission m has a release time r_m , absolute deadline d_m , origin, destination, service type, priority p_m , state and configured penalties. An aircraft has a role, assignment status, position, remaining endurance, landing compatibility and commandability.

A disruption can alter ground travel time, interrupt a selected air chain, remove an aircraft from supervisory control or eliminate a landing option. The same external event can affect different policies differently because their earlier assignments determine the mission’s exposure. Cross-modal reconfiguration changes a task’s assigned resource or transport mode, or changes the continuation of an existing operation. It can improve an emergency’s outcome while changing the service state of an incumbent mission.

The supervisor receives an observation

$$\hat{S}_t = \mathcal{O}_t(S_{0:t}), \quad \tau(t) \leq t, \quad (2)$$

where $\mathcal{O}_t$ incorporates snapshot sampling and event ordering and $\tau(t)$ is the snapshot timestamp. The age $t - \tau(t)$ is useful but does not completely describe visibility: an event injected after a snapshot at the same simulation second is absent from that snapshot. This distinction becomes operationally important in the observation-scheduling experiment.

3.2. *Candidate information and feasible alternatives*

A deterministic mapping Φ constructs the candidate information interface,

$$\mathcal{I}_t = \Phi(\hat{S}_t), \quad \mathcal{C}_{m,t} = \{c \in \mathcal{I}_{m,t} : \text{legal}(c \mid \hat{S}_t) = 1\}. \quad (3)$$

$\mathcal{I}_{m,t}$ contains the records associated with mission m , whereas $\mathcal{C}_{m,t}$ denotes the sub-

set marked legal. The distinction is necessary because the interface retains rejected resource records and their rejection reasons. A candidate record identifies the mission and resource, transport mode, implied action, estimated remaining completion duration $\hat{\eta}_c$, feasibility status, endurance information, destination conditions and any incumbent service that would be displaced.

Air feasibility depends on availability, mission compatibility, commandability, battery/endurance conditions, landing-site compatibility and modeled restrictions. Reassignment of an occupied aircraft requires an explicitly reassignable incumbent mission with strictly lower priority than the target. A ground candidate is available when the mission permits ground fallback and a travel-time estimate is present. These candidates represent the alternatives generated by the implemented service model; they do not enumerate every possible route or future schedule.

The interface contains no B2 objective score, optimizer selection or manager-specific recommended action. Nevertheless, it embodies shared modeling choices, including priority constraints, travel-time approximations and the organization of multi-mission records. Manager-agnostic therefore means that policies share the same generation rules and facts at a given state, rather than that the representation is free of operational assumptions. Once policies take different actions, their subsequent states and candidate sets can legitimately differ.

3.3. *Supervisory actions and execution-time constraints*

At a decision epoch, policy k proposes one structured intervention:

$$a_t^{\text{prop}} = \pi_k(\hat{S}_t, \mathcal{I}_t), \quad k \in \{\text{B0}, \text{B1}, \text{B2}, \text{B4b}\}. \quad (4)$$

The action contract includes assignment and reassignment actions, route or destination changes, ground fallback, and management responses such as delay or no action. A proposal need not correspond to selecting an aircraft. In particular, B0 restricts cross-modal emergency support to the ground alternative.

Let $t_e = t + \delta_{\text{exec}}$ be a proposal's scheduled execution time. Let $Q(a, t_e)$ indicate that the proposal remains the active command version for the relevant task, and let $\Gamma(a, S_{t_e})$ denote the semantic and feasibility check using the current execution state. Abstracting the action-specific normalization performed by the implementation, the executed intervention is

$$u_{t_e} = \begin{cases} \text{Executor}(a_t^{\text{prop}}, S_{t_e}), & Q(a_t^{\text{prop}}, t_e) = 1 \wedge \Gamma(a_t^{\text{prop}}, S_{t_e}) = 1, \\ \emptyset, & \text{otherwise.} \end{cases} \quad (5)$$

A proposal replaced by a newer command is superseded; a proposal that fails the current checks is rejected. Neither event is equivalent to completed transport. The subsequent state follows

$$S_{t+1} = F(S_t, u_t, e_t, \ell_t), \quad (6)$$

where e_t represents exogenous events and ℓ_t represents local contingency responses.

This notation describes the implemented supervisory interface and event-driven simulator evolution.

3.4. *Service outcomes and temporal feasibility*

For a completed mission, completion duration and deadline violation are

$$T_m = c_m - r_m, \quad V_m = \mathbf{1}[c_m > d_m], \quad (7)$$

where c_m is the absolute completion time. Completion exactly at the deadline is on time. A useful diagnostic for a candidate at decision time t is its estimated temporal margin,

$$\sigma_m(c, t) = d_m - t - \delta_{\text{exec}} - \hat{\eta}_c, \quad \mathcal{C}_{m,t}^{\text{timely}} = \{c \in \mathcal{C}_{m,t} : \sigma_m(c, t) \geq 0\}. \quad (8)$$

A nonempty feasible set does not ensure a nonempty set of alternatives predicted to finish on time. Even a nonnegative estimated margin does not guarantee on-time completion, because subsequent failures, queue changes and estimation error can alter the outcome. Equation (8) connects candidate facts to the decision regimes analyzed in Figure 4.

We evaluate service resilience through path restoration, deadline compliance, incumbent-service preservation and priority-weighted loss. These outcomes describe complementary aspects of supervisory recovery. They do not measure infrastructure repair or imply that all parts of the transport system have returned to their original operating conditions.

4. *Ground–Low-Altitude Mobility Manager*

4.1. *Global state and closed-loop environment*

The framework couples SUMO for road traffic with BlueSky for aircraft evolution. (Alvarez Lopez et al. 2018), (Hoekstra and Ellerbroek 2016) A master orchestrator advances both layers in 1 s steps, maintains the service registry and applies events and accepted interventions. Figure 1(b) places the replaceable policy within this common sequence. Local aircraft contingencies remain independent of supervisory inference.

The snapshot combines road estimates and closures with aircraft assignments, endurance, commandability, landing-site conditions and mission priorities and deadlines. Ground candidates use the shared D1–H1 corridor ETA, including for eligible missions represented in the multi-mission interface. When ground fallback executes, the implementation queries the current SUMO corridor estimate and schedules completion using the rounded duration. Completion is recorded by a mission timer; fallback does not simulate a separate dispatched road vehicle and trajectory for each mission.

Coupling occurs through task status and resource assignment. A new air assignment follows the resource’s current position to the mission origin and destination; an interrupted-mission candidate uses the contingency site V3 before the destination. Pre-emption detaches the incumbent mission and changes the resource’s assignment. Local return or diversion evolves separately, but replacement dispatch is not gated by verified

arrival of the original payload at V3. The registry has no payload-custody, unloading, reloading or re-shipment process. Ground fallback changes mode and schedules completion from the live corridor ETA, with the proposal ETA used only if that query is unavailable. These semantics represent service replacement, with physical transfer time examined separately through an offline margin sensitivity in Supplementary Section S4.

Algorithm 1. Ground–air supervisory service reconfiguration

- (1) Initialize both simulators, the mission/resource registry and the observation and pending-command stores.
 - (2) At each simulation step, apply scheduled events. Their local contingency responses run independently of supervisory inference.
 - (3) At an event or periodic decision opportunity, observe the joint state (the latest stored snapshot in E4A); build candidate facts for the actionable mission(s).
 - (4) Select one supervisory intervention using policy π ; normalize missing route/target fields and validate its structure, semantics and modeled feasibility. Log rejected proposals.
 - (5) Issue an accepted intervention immediately, or queue it for $t + \delta_{\text{exec}}$. Keep the original due time for an identical pending command; a changed command supersedes the previous one for that mission.
 - (6) After same-time events and their decisions, process due commands: normalize again, refresh state and revalidate. Skip completed effects; on stale rejection request a new decision; otherwise execute.
 - (7) Advance SUMO and BlueSky by 1 s; update mission/resource states, ground completion timers and scheduled observations. Repeat until the horizon.
-

Algorithm 1 follows the implemented event-before-pending order. Ordinary periodic decisions are suppressed while an actionable mission already has a pending command; E4A instead follows its scheduled polling protocol. The identical-command rule is the queue correction evaluated in Section 6.6.2; independent local contingencies do not wait for that queue.

4.2. *Candidate generator and shared executable interface*

The candidate generator maps the observed global state into a shared executable candidate interface. Generation computes the candidate facts; the interface specifies the fields and action semantics available to the policy. Together they make air and ground alternatives comparable without prescribing a selection. Remaining completion estimates use a common duration convention. Air estimates combine the modeled route geometry, role-specific cruise speeds and dispatch overhead. The interface also reports absolute estimated completion time, the remaining deadline margin and structural consequences of preemption. Ground uses the common fallback estimate. These estimates support supervisory choice. Execution refreshes operational quantities from the environment, while deterministic checks apply the modeled constraints.

The generator does not run B2 or receive its preference weights. B1 and B2 calculate their own selections from the shared table, while B4b receives the same table within the prompt. A structural interruption indicator identifies whether reassignment displaces an existing service; it is not a learned welfare estimate or a B2-weighted cost.

E2 and E4 add failure and risk-zone facts to the E1 interface. E3 extends it to

Table 1. Main information groups in the executable candidate interface.

Group	Information supplied	Supervisory role
Target mission	Identifier, priority, origin, destination, deadline and state	Establish task requirements and urgency
Resource and action	Aircraft or ground identifier, mode, dispatch/reassignment/fallback action	Connect the alternative to the execution contract
Feasibility	Legal flag, rejection reason, compatibility, commandability and destination state	Distinguish available interventions from rejected alternatives
Completion and time	Remaining ETA, estimated absolute completion, deadline comparison	Compare transport duration and remaining margin
Existing service	Preempted mission identifier and priority, interruption indicator	Make the consequence of reassignment visible
Air operating state	Battery/endurance information and failure-related restrictions	Represent the modeled operating conditions

multiple actionable missions organized by priority and deadline. Information derivation and execution semantics are shared across policies within each experiment.

A decision with service-preservation pressure. At Site A, high disruption and high medical workload leave a timely ground route and one faster, preemptive air option. At 300 s in seed 20240601, the HIGH-priority emergency is due at 600 s. The logged candidates are:

Resource / mode	ETA (s)	Margin (s)	Legality	Incumbent	Preemption consequence
GROUND / ground	228.617	71.383	Legal	None	None
M-UAV-02 / air	212.823	87.177	Legal	M-L-002, NORMAL	Interrupt incumbent service
EVTOL-01, L-UAV-01, M-UAV-01 / air	—	—	Illegal	Occupied	Non-reassignable

B2 selected ground and completed in 229 s; B4b reassigned M-UAV-02 and completed in 212 s, leaving one damaged incumbent service. B1 also selects the legal air option under its air-first rule. The same candidate facts support all three policies. The estimated 15.794 s air advantage improves speed, but both alternatives meet the emergency deadline and only air displaces an incumbent. Thus the fastest option can have the larger system service cost. Candidate ETA is an estimate; realized air completion also reflects simulator motion and the arrival radius.

4.3. Supervisory policies

The four primary policies are summarized in Table 2. All use the common action contract and execution constraints. B0 provides a reference for emergency transport without supervisory air support, while B1, B2 and B4b compare alternative coordinated policies.

Background air services remain present under B0.

Table 2. Supervisory policies and their comparison roles.

Policy	Selection rule	Interpretation
B0: ground-only	Serves the eligible target by ground fallback, without retasking an air asset for cross-modal support	Reference for the total effect of enabling coordination
B1: air-first rule	Chooses the legal air candidate with minimum ETA, using endurance to break ties; considers ground when no legal air option remains	Simple deterministic coordination
B2: fixed-objective heuristic	Selects a target under the priority rule and minimizes a fixed scalar score over legal air and ground candidates	Transparent speed–service trade-off
B4b: LLM-candidate	Reads the shared state and explicit candidate table and returns a structured intervention	Language-based implementation of the policy layer
B4a: interface ablation	Uses the same prompt template and checker as B4b with the explicit candidate section removed	Additional E1 interface ablation only

B1’s air-first ordering is important: it minimizes ETA among legal air candidates, rather than minimizing travel time over both modes. B2 is a fixed-objective heuristic that ranks legal air and ground candidates for the priority-selected actionable mission using

$$J_t(c) = \hat{\eta}_c + 10 \max(0, t + \hat{\eta}_c - d_m) + \mathbf{1}_{\text{preempt}}(c)(\kappa_{p(c)} + 30) + 10\rho_c, \quad (9)$$

where $\kappa_{\text{LOW,NORMAL,HIGH,CRITICAL}} = (20, 40, 80, 100000)$ and $\rho_c = \mathbf{1}[b_c < 120] + \mathbf{1}[b_c < 60]$ for endurance margin b_c in seconds. Ground candidates incur neither preemption nor endurance penalties. The score uses configured seconds-equivalent trade-offs and contains no separate ground-delay term. Equal- or higher-priority preemption is already excluded by the feasibility rules. The weights define a fixed speed–service-preservation preference and remain unchanged across comparisons.

B4b uses the common prompt template with temperature 0, JSON output mode and an output-token budget of 8192. The recorded backend identifier is `corp-ai/openai/deepseek-v4-pro`; it identifies the configured service, with the underlying release unverified. B4a retains the same snapshot, prompt rules and checker but removes the candidate section. This removes per-resource air ETA, explicit legal/rejection flags, deadline consequences and the mapped preempted-service consequence. Raw assignments, priorities, positions and ground ETA remain available. We therefore define the ablation as *explicit candidate-information support*. Supplementary Section S2 supplies the field comparison and full prompt file. Recorded errors and structured retries remain in the analyzed cohort.

4.4. *Deterministic checking*

The validation pipeline checks action structure, identifiers, mission and aircraft status, commandability, compatibility, destination availability, priority restrictions and the relevant modeled operating constraints. The distinction between preselection filtering and action checking is deliberate. Candidate legality describes the observed state; execution checks address whether the selected intervention can be applied to the state that exists when it is due.

Rejected proposals are logged rather than silently converted into a preferred action. An accepted assignment changes the registry and simulator commands through deterministic execution rules. These rules apply to every supervisory implementation. The execution layer supplies modeled feasibility enforcement for every policy; operational aviation certification lies outside this service model.

4.5. *Execution, pending actions and local contingencies*

The executor maintains the distinction between proposal, issue, execution and transport completion. In the execution-delay experiment, accepted proposals enter a pending queue. Task-specific versioning allows a changed command to supersede an earlier pending command; an identical normalized command retains the original due time, and the execution pipeline revalidates the current action before application. A change in the environment can therefore alter both when a command executes and which command ultimately survives.

Local aircraft contingency behavior continues independently of supervisory decisions. In the modeled command-link-loss case, for example, an affected aircraft becomes unavailable for further supervisory commands and follows its local return procedure. The manager can then select another air resource or ground fallback for the interrupted service. Service recovery thus combines a common local response with a potentially policy-dependent replacement decision.

4.6. *Decision opportunities and observation scheduling*

New or interrupted missions can trigger supervisory decisions. Outside the observation-interval experiment, periodic decisions are also enabled every 30 s while a task is waiting, requires replanning or is interrupted. In E3, the condition covers actionable missions in the registry, including background services. Each accepted intervention is followed by continued simulator evolution and subsequent observations.

E4A replaces this ordinary observation arrangement with controlled snapshot refresh and polling schedules. The policy uses the latest observed snapshot rather than an unrestricted view of current state. E4B instead introduces controlled execution waiting, and E4C adds inert input records around the unchanged core service problem. These experiments isolate different implementation conditions while preserving the respective policy definitions.

5. Experimental design

5.1. Study sites and modeled services

The three study areas are based on OpenStreetMap road data from Suzhou, China (A), Amsterdam, the Netherlands (B), and Edmonton, Canada (C). OpenStreetMap supplies the geographic network substrate. (Haklay and Weber 2008) The configured study-area extent is $3.2 \text{ km} \times 3.2 \text{ km}$ at each site. Figure 2 shows the imported road substrates and representative service connections. Network conversion can retain geometry beyond the nominal area, so the figure uses a common projected scale and shows the full imported substrates rather than presenting their outlines as identical nominal-area boundaries.

Contrasting network morphology

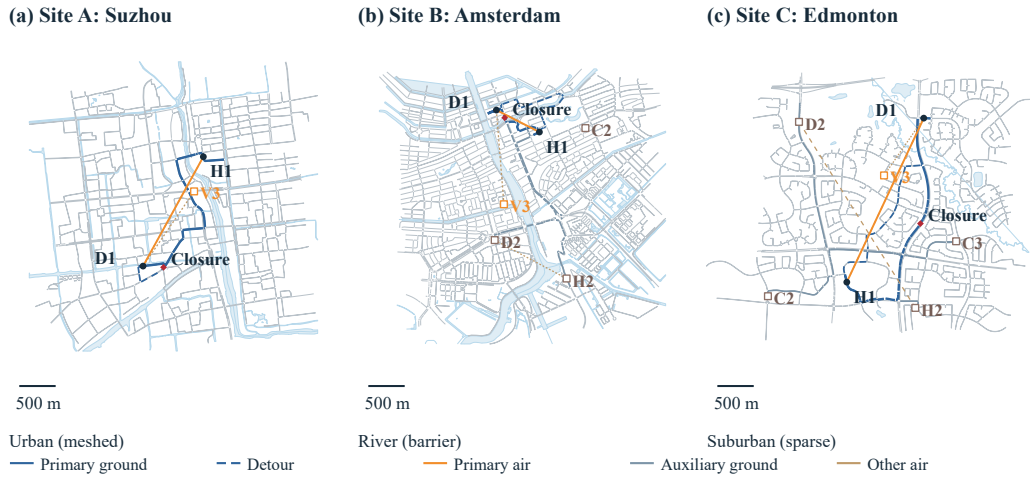


Figure 2. Three contrasting network morphology contexts, drawn from archived projected road and facility geometry. Blue indicates the primary ground route and dashed blue the detour; orange denotes low-altitude connections. Thin blue-grey routes show auxiliary ground ODs, and muted orange dashed lines show background air connections where present. D1/H1 mark reference endpoints, V3 a backup landing site, and D2/H2/C2/C3 auxiliary facilities. Co-located landing nodes are not labeled separately. Closure marks the reference disrupted road link. Auxiliary routes describe site context, not additional E1 missions. Maps use a common projected scale and 500 m scale bars. Road and water data: © OpenStreetMap contributors. Facilities, air operations and disturbances are modeled.

Road length is the sum of directional passenger-edge lengths apportioned by the fraction of each polyline inside the configured study polygon. Both density denominators use that polygon’s projected area, approximately 10.24 km^2 . Intersection counts include only nodes inside it, with at least three distinct neighbors in the full graph. The full routing network retains outside roads; its topology and fixed 200-pair OD sample are explicitly labeled separately. D1–H1 quantities describe the reference service pair. Supplementary Section S4 gives the boundary correction.

The sites offer contrasting operating contexts. Site A has a meshed urban network with meaningful choices between continued ground service and occupied air resources. Site B combines the highest within-boundary road density with water-barrier constraints on the reference ground corridor. Site C has a higher dead-end ratio and mean sampled OD circuitry, with a longer reference service distance. These characteristics supply contrasting transport alternatives; their causal separation is discussed in Section 7.

Table 3. Network and reference-service descriptors.

Descriptor	A: Suzhou	B: Amsterdam	C: Edmonton
Statistical polygon area (km ²)	10.2450	10.2361	10.2383
Within-boundary road length (km)	181.934	217.946	168.410
Within-boundary road density (km/km ²)	17.758	21.292	16.449
Within-boundary intersection density (km ⁻²)	37.970	100.331	39.655
Full routing-network nodes	542	1877	791
Full directional passenger edges	1286	3514	1681
Full-network mean node degree	2.830	2.663	2.420
Full-network dead-end ratio	.1845	.1193	.2882
Full-network sampled OD circuitry	1.1969	1.1409	1.4721
D1-H1 network distance (km)	2.9222	1.1192	3.4296
D1-H1 Euclidean distance (km)	1.7403	.7246	2.5285
D1-H1 circuitry	1.6791	1.5446	1.3564
Reference closure ETA increase (%)	37.87	88.69	34.76

Every site uses a four-aircraft core fleet: two medical UAVs, one logistics UAV and one passenger eVTOL. Compatibility restricts which aircraft can serve a given mission. Low medical workload leaves the medical aircraft idle, whereas high workload assigns them to existing services. Facilities, aircraft operations, landing sites and failures are scenario constructs. These are modeled services on real road substrates.

5.2. *Shared controls and site adaptations*

Policies share mission semantics, the action contract, candidate-generation rules and deterministic execution constraints within an experiment. The prompt and B2 objective remain specified. Geographic coordinates, road alternatives, aircraft travel distances and failure placement vary across sites. Failure times are adapted to intercept the modeled air operation, rather than applying an identical time that would affect different flight phases. Thus, the cross-site evaluation tests the same supervisory architecture in different network and service contexts.

Two comparisons must be distinguished. B0 versus a coordinated policy measures the total difference associated with allowing cross-modal support, including a difference in available transport modes. B1 versus B2 versus B4b compares supervisory implementations under common information and action constraints. In contrast, B4a versus B4b tests explicit candidate presentation within the same LLM implementation. These comparisons answer different questions and are not combined into one ranking of algorithmic intelligence.

5.3. *Disturbance experiments*

E1: ground disruption and selective support. A new blood-transport mission and a ground disruption occur at 300 s. Ground disruption has three levels, urgency has HIGH and CRITICAL levels, and medical workload is low or high. HIGH urgency has 300 s of deadline slack and CRITICAL urgency has 180 s. The resulting 12 scenarios run for 900 s. Six scenarios per site provide the interface ablation, with ten matched seeds each. The 180 extra runs are B4a runs; their B4b counterparts belong to the primary cohort.

E2: individual air-system failures. The primary CRITICAL mission is released at

Table 4. Experiment design and sample sizes. Each primary scenario–arm–policy cell contains 20 paired seeds.

Experiment	Factors	Sites	Primary runs	Additional B4a runs
E1	3 ground disruption levels × 2 urgency levels × 2 workloads	A, B, C	2,880	180
E2	16 feasible failure-family/context combinations	A, B, C	3,840	0
E3 v2	12 scenarios across four disturbance/task levels	A, B, C	2,880	0
E4A	Observation intervals of 10, 30, 60, 120, 300 s	A	400	0
E4B	Execution delays of 0, 1, 5, 10, 20, 30, 60 s	A	560	0
E4C	Total aircraft records of 5, 10, 20, 30, 50	A	400	0
Total			10,960	180

300 s and has deadline 480 s. Six failure families model command-and-control link loss, GNSS degradation, UTM degradation/outage, landing-site failure, unknown-aircraft intrusion and flyaway behavior with a moving risk envelope. Context C1 affects a peripheral part of the operation, C2 interrupts the primary service chain while retaining another air option, and C3 removes the relevant air alternatives. F3–C2 and F4–C2 are excluded because the specified mechanisms cannot create the intended combination of interruption and surviving air alternative. The resulting 16 scenarios use fault times of 360, 322 and 383 s at A, B and C, respectively.

E3: compound disturbance and competing services. Twelve scenarios form four levels. L1 comprises two single-failure, single-emergency anchors. L2 comprises four two-failure, single-emergency cascades. L3 comprises four single-failure scenarios with two simultaneous emergency missions and tighter medical-resource availability. L4 comprises two scenarios with multiple emergencies and two or three failures. In the two-emergency scenarios, the second medical mission is released at 300 s with HIGH priority and 240 s of deadline slack. The primary CRITICAL and secondary HIGH missions have delay penalties of 60 and 45 and cancellation penalties of 600 and 450, respectively. Horizons are 900 or 1,200 s. These levels change both disturbance structure and task demand and are not a one-dimensional ladder of difficulty.

5.4. *Observation, execution delay and input burden*

E4 retains Site A’s core task and separates three operational changes. E4A varies the snapshot observation interval, with a fault at 371 s. Scheduled polling continues over the observation experiment’s horizon, including after some transport decisions have resolved. E4B introduces the same specified decision-to-execution delay for every pol-

icy, with a fault at 360 s. E4C enlarges the surrounding input using unavailable, non-commandable aircraft records with zero endurance and incompatible landing capabilities, together with inert facility and completed-mission records.

E4C does not add usable transport capacity. Its core fleet remains four aircraft, and candidate invariance is checked at matched decision stages. The experiment therefore tests input burden and selection quality for a fixed transport decision. It does not test scheduling fifty operational aircraft or serving a proportionally larger set of urgent missions.

5.5. Outcomes, pairing and statistical interpretation

The primary seeds are 20240601–20240620. Each controls the SUMO random seed and a common initial-state realization, including progress and endurance perturbations for aircraft. Busy-aircraft progress is sampled along its route, and remaining endurance receives a seed-dependent reduction of up to 15%. Comparisons pair policies by scenario and seed. Some outcomes are constant within a cell because the perturbations do not change the executed action or resulting completion time.

Emergency completion and deadline violation follow Equation (7). Existing-service damage counts non-primary missions whose horizon state is INTERRUPTED, NEEDS_REPLAN, CANCELLED or FAILED. It measures final service status rather than cumulative historical interruptions or physical damage. Issued actions identify air intervention; rejected proposals do not count as interventions.

For recovery, let $A_i = 1$ if a run’s primary air chain is actually affected and $R_i = 1$ if an executable replacement service is established. Conditional path restoration is

$$\hat{P}_{\text{rec}} = \frac{\sum_i A_i R_i}{\sum_i A_i}. \quad (10)$$

When no run is affected, the result is N/A. Pairwise recovery-time comparisons include only pairs in which both policies are affected. Completion and deadline comparisons retain all matched runs. Failure-to-replan latency uses the first issued mission-specific DISPATCH, REASSIGN, DIVERT, REROUTE or GROUND_FALLBACK action after the fault, excluding NO_ACTION. Path restoration and completed transport are separate events.

E3’s system weighted loss (SWL) is

$$L_H = \sum_{m \in M_H} w_{p_m} \ell_m(H), \quad w_{\text{CRITICAL,HIGH,NORMAL,LOW}} = (4, 3, 2, 1), \quad (11)$$

with

$$\ell_m(H) = \begin{cases} q_m^{\text{delay}}, & m \text{ completed and } c_m > d_m, \\ q_m^{\text{cancel}}, & \text{state}_m(H) \in \mathcal{U}, \\ 0, & \text{otherwise,} \end{cases} \quad (12)$$

$$\mathcal{U} = \{\text{CANCELLED, FAILED, INTERRUPTED, NEEDS_REPLAN, WAITING}\}.$$

Late completion incurs a fixed penalty, not a charge per second late. EN_ROUTE and ASSIGNED services incur no horizon penalty. A terminal-registry audit found all 4,320 E3 task instances with deadlines at or before the horizon completed; none were omitted as overdue ongoing tasks. A late primary CRITICAL mission therefore contributes 240 units and a late secondary HIGH mission contributes 135 units. SWL is expressed in configured loss units, not monetary welfare. We also report the CRITICAL+HIGH component when comparing compound-level policies.

The fixed scenario panel defines the estimand. Equal weights are assigned to scenario-seed runs, preserving the E3 level sample sizes of 40, 80, 80 and 40 per policy. The initialization code restarts `random.Random(seed)` in each scenario, traverses aircraft in sorted order and also passes that seed to SUMO. Scenario workloads can change the number of draws, but do not establish independent streams. We therefore group the complete fixed panel by seed before estimating uncertainty and pair policies within each scenario and seed.

For a complete panel with J scenarios and K seeds, define

$$D_s = \frac{1}{J} \sum_{j=1}^J (Y_{js}^{(u)} - Y_{js}^{(v)}), \quad \bar{D} = \frac{1}{K} \sum_s D_s, \quad \text{CI}_{95\%} = \bar{D} \pm t_{0.975, K-1} \frac{s_D}{\sqrt{K}}. \quad (13)$$

Primary panels use $K = 20$; the six-scenario E1 ablation uses $K = 10$. Means and scenario weights are unchanged by blocking. Continuous comparisons use a paired t test on seed aggregates, with the previously specified signed-rank fallback for constant nonzero differences. Identical differences receive $p = 1$. Binary comparisons swap policy labels for entire seed blocks in an exact paired randomization test, preserving within-seed dependence. This replaces run-independent McNemar inference for multi-scenario panels. Unadjusted intervals describe fixed-panel seed uncertainty; a zero-width interval means no variation across these seed aggregates. Supplementary Section S3 specifies the conditional-denominator calculation and provides the complete revised tests.

Holm adjustment retains the original comparison families: three E1 ablation endpoints per site; 12 scenario-specific E1 completion tests; 12 E2 endpoints per site; and 12 E3 compound comparisons per site. E4 retains separate per-policy, per-metric non-anchor comparisons with OBS30, D00 and N05. Adjusted $p < 0.05$ is the decision criterion. (Holm 1979) The sites form separate fixed transport contexts, and nonsignificance is not statistical equivalence.

5.6. Provenance and model scope

Experiments evaluate supervisory service replacement with four core aircraft, specified fault state machines and the ground timer abstraction described in Section 4.1. External inference calls do not advance physical simulation time. E4A changes observed information and E4B inserts a controlled execution wait; wall-clock backend latency is recorded separately. The numerical revision preserves original runs, corrects the B0 D60 duplicate-command cell through a local implementation overlay, and recalculates fixed-panel statistics. Supplementary materials provide parameters, prompts, analysis scripts and revision provenance.

6. Results

6.1. Selective air support and incumbent-service cost

Enabling air support improved emergency transport, while coordinated policies imposed different incumbent-service costs (Figure 3). At A, B2 and B4b reduced deadline violations from B0's 33.33% to 16.67%. Their mean completion durations were 147.62 and 146.63 s, respectively, but B4b incurred 0.133 damaged services per run against B2's 0.058. Supplementary Table S1 retains the complete site-policy summaries and intervals.

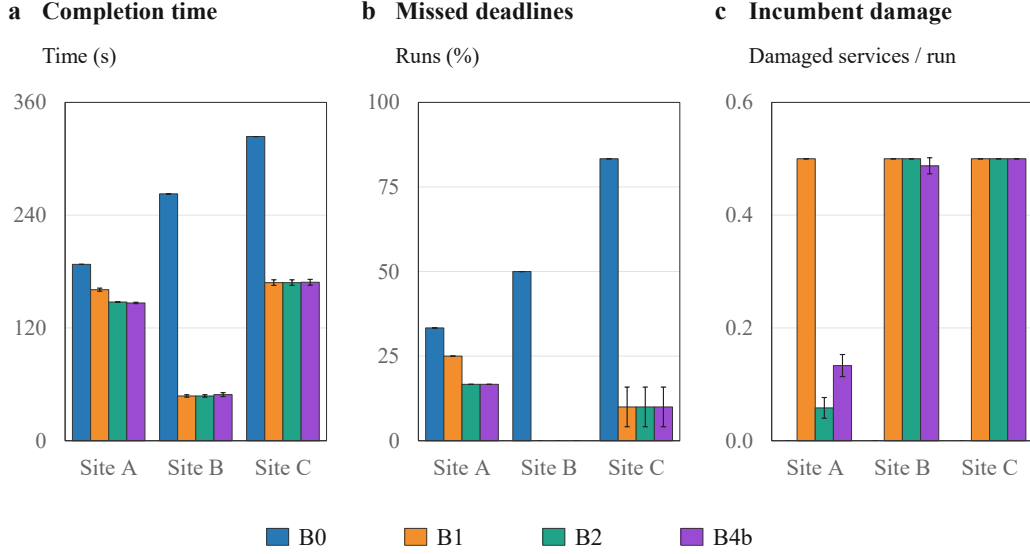


Figure 3. E1 service benefit and incumbent cost. Clustered bars show site-policy mean completion time, missed deadlines and incumbent services damaged at the horizon. Whiskers are 95% seed-block intervals (240 runs, 20 seed blocks per site and policy). All bars start at zero; zero-width intervals have coincident caps. Colors identify policies consistently across the result figures. Supplementary Table S1 retains exact values.

The high-disruption, HIGH-urgency, high-workload scenario at A exposes this trade-off. Ground could meet the 300 s slack, and B2 selected it in all 20 runs. B4b reassigned an occupied aircraft in 12 runs and used ground in eight. Relative to B2, completion decreased by 11.95 s (95% CI -17.330 to -6.570; Holm-adjusted $p = 0.002100$), while damage increased by 0.600 services per run. Both policies met every emergency deadline. The speed gain therefore came from preempting a service even though ground retained a positive deadline margin. A manager must price that displacement alongside emergency speed.

At B and C, large gains over ground-only operation coexisted with similar outcomes among coordinated policies. Their deadline-violation rates were respectively 0% and 10%, against B0's 50.00% and 83.33%. These differences distinguish the benefit of enabling a useful transport mode from the effect of selecting a more complex supervisor.

6.2. The candidate interface shapes supervisory choices

Explicit candidate-information support improved both speed and service preservation at A (Table 5). B4b reduced completion duration, incumbent-service damage and air

intervention relative to B4a. Completion decreased by 13.567 s, damage by 0.383 services per run and air intervention by 38.333 percentage points. These jointly favorable changes indicate more selective use of the same transport resources.

Supervisory decision regimes

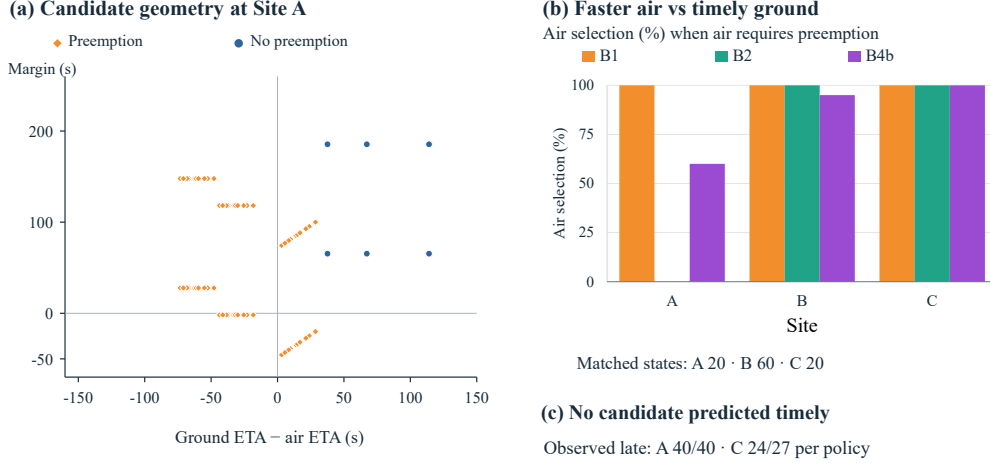


Figure 4. Supervisory decision regimes from E1 first-decision logs. (a) Site A candidate geometry relates the air ETA advantage to the best remaining deadline margin; symbols distinguish preemption. Repeated locations represent multiple matched states. (b) Air selections when faster air requires preemption and ground remains timely, with 20, 60 and 20 states at A, B and C. (c) Among states with no candidate predicted timely, observed late counts are 40/40 at A and 24/27 at C per policy; B has no such states. First candidate snapshots match across B1, B2 and B4b in all 720 scenario-seed states. Subsequent actions and trajectories may differ. ETA-based groups are descriptive, not a new policy score.

Table 5. Paired E1 interface ablation: B4b minus B4a.

Site	Endpoint	Pairs	Difference	95% seed-block interval	Holm p
A	Completion (s)	60	-13.567	[-15.978, -11.155]	9.33e-07
A	Damaged services/run	60	-0.383	[-0.441, -0.326]	3.27e-07
A	Air intervention (pp)	60	-38.333	[-44.093, -32.574]	0.001953
B	Completion (s)	60	1.450	[-2.288, 5.188]	1.000
B	Damaged services/run	60	-0.017	[-0.054, 0.021]	1.000
B	Air intervention (pp)	60	-1.667	[-5.437, 2.104]	1.000
C	Completion (s)	60	0.000	[0.000, 0.000]	1.000
C	Damaged services/run	60	0.000	[0.000, 0.000]	1.000
C	Air intervention (pp)	60	0.000	[0.000, 0.000]	1.000

Each site retains a three-endpoint adjustment family. Intervals use ten seed blocks; air intervention is a risk difference in percentage points (pp) with exact whole-block label-swap inference.

The information-support effect was site-dependent: B had no difference passing Holm adjustment and C had zero difference on all three endpoints. Figure 4 relates choices to ETA advantage, deadline margin and preemption. When ground is both faster and non-preemptive, it dominates those two service criteria; B1 can still prefer air because its rule ranks only air candidates. This explains why candidate availability alone does not ensure a useful selection. At A, 20 states offer faster preemptive air and timely ground: B1, B2 and B4b choose air in 20, zero and 12 cases. At B and C, the corresponding counts are 60/60/57 and 20/20/20, showing that a speed-service conflict does not inevitably produce policy divergence. When A has no predicted timely candidate, all 40 runs per coordinated policy are late despite different mode choices. At C, 24 of 27 such cases are late; the remaining three expose the approximation in candidate ETA. These observations connect explicit information support to decisions without equating predicted margins with guaranteed completion.

6.3. *Restored paths do not guarantee timely recovery*

Every affected coordinated run established an executable replacement path in E2 (Supplementary Figure S1). Prior assignments changed exposure, so B4b’s affected counts differed from B1/B2’s; B0’s conditional recovery was N/A. Despite complete restoration among affected runs, B4b’s all-run deadline-violation rates remained 39.06%, 37.50% and 62.81% at A, B and C. A surviving route could take longer than the remaining deadline margin.

B4b completed later than B2 at A, with an adjusted difference passing the specified test (Table 6). Its conditional recovery-time difference did not pass correction. B and C completion differences were smaller and also did not pass Holm adjustment. Rapidly establishing replacement service does not remove the travel time that remains after the fault; dispatch latency and deadline performance therefore answer different operational questions.

Table 6. Selected E2 paired comparisons: B4b minus B2.

Site	Endpoint	Pairs	Difference	95% seed-block interval	Holm p
A	Completion (s)	320	4.525	[2.412, 6.638]	0.003069
A	Recovery time (s)	191	3.927	[0.930, 6.924]	0.142465
B	Completion (s)	320	0.175	[0.060, 0.290]	0.055710
B	Recovery time (s)	199	0.281	[0.098, 0.465]	0.055710
C	Completion (s)	320	0.922	[-0.214, 2.057]	1.000
C	Recovery time (s)	200	0.255	[-0.138, 0.648]	1.000

Adjustment uses the original 12-endpoint family per site. Completion uses all matched runs; recovery time uses jointly affected pairs. Intervals are unadjusted and p-values are multiplicity-adjusted. At C, 201 B4b runs were affected but only 200 pairs were jointly affected. Backend errors and retries remain included; some deterministic replacement decisions occur within the fault’s simulation second.

6.4. *Compound service loss depends on surviving options*

Coordination reduced aggregate priority-weighted loss in E3 (Figure 5; Supplementary Tables S3 and S7). At A, SWL fell from 240 under B0 to 100 under B1/B2 and 102.125

under B4b. At B and C, coordinated policies shared aggregate means of 127.5 and 227.5, respectively, against 307.5 under B0.

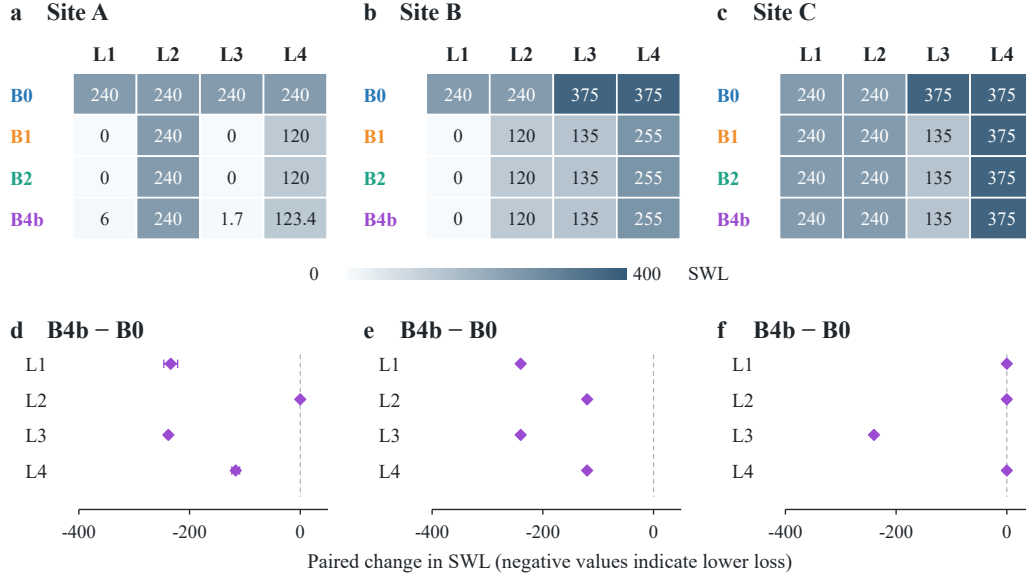


Figure 5. E3 service loss and coordination benefit. (a–c) All policy-level means share one SWL color scale; noninteger labels are rounded to one decimal. Each policy has 40 runs in L1/L4 and 80 in L2/L3. (d–f) Paired B4b-minus-B0 differences and 95% seed-block intervals; negative values indicate lower loss under B4b. Levels jointly vary disturbance and demand. Exact values remain in Supplementary Tables S3 and S7.

These averages contained clear limits. At A, all policies retained loss 240 in L2. At C, coordination produced no SWL reduction in L1, L2 or L4; its improvement was confined to L3. Thus, the E1 benefit of air availability did not persist across every combination of failures and service demand.

B4b did not significantly reduce SWL or CRITICAL+HIGH loss relative to B1 or B2 in the recorded compound comparison families. At A, its L3 and L4 SWL differences were +1.6875 (95% CI [-1.844, 5.219]) and +3.375 ([-3.689, 10.439]), with adjusted $p = 1$. The corresponding compound-level differences at B and C were zero. The transport benefit follows the surviving deadline-compatible alternatives, while policy differences contribute little additional protection in these panels.

Primary speed alone also missed service benefits. At A, coordinated policies reduced SWL despite longer mean primary completion than B0. The configured loss depends on which priority-weighted services cross deadlines, not solely on average primary travel time.

6.5. Contrasting morphology and policy sensitivity

The sites separate coordination opportunity from sensitivity to policy. A contains decisions where ground remains timely and air reassignment offers a smaller speed gain at an incumbent-service cost. It also shows the strongest interface effect. B combines a dense network with a barrier-constrained reference corridor; C has more dead ends, higher sampled OD circuitry and a longer reference trip. Both show large E1 coordination benefits with similar coordinated-policy outcomes.

C's limited E3 gains qualify this pattern: useful alternatives must preserve the rel-

event deadline to avoid the configured penalty. Across sites, road geometry, facilities, air travel times and fault timing change together. Morphology contextualizes the manager’s choices; this comparison does not estimate an independent causal effect of density, circuitry or redundancy.

6.6. *Timing conditions and input burden*

6.6.1. *Observation scheduling*

Longer observation intervals could exhaust the available service margin (Figure 6a; Supplementary Table S4). B2/B4b completion increased from 140 s at OBS10 to 482 s at OBS300. Their results coincided at every arm; B1 differed at OBS300, completing in 521 s. All B4b OBS10–OBS60 runs met the deadline, with OBS60 exactly on time, whereas OBS120/OBS300 were all late.

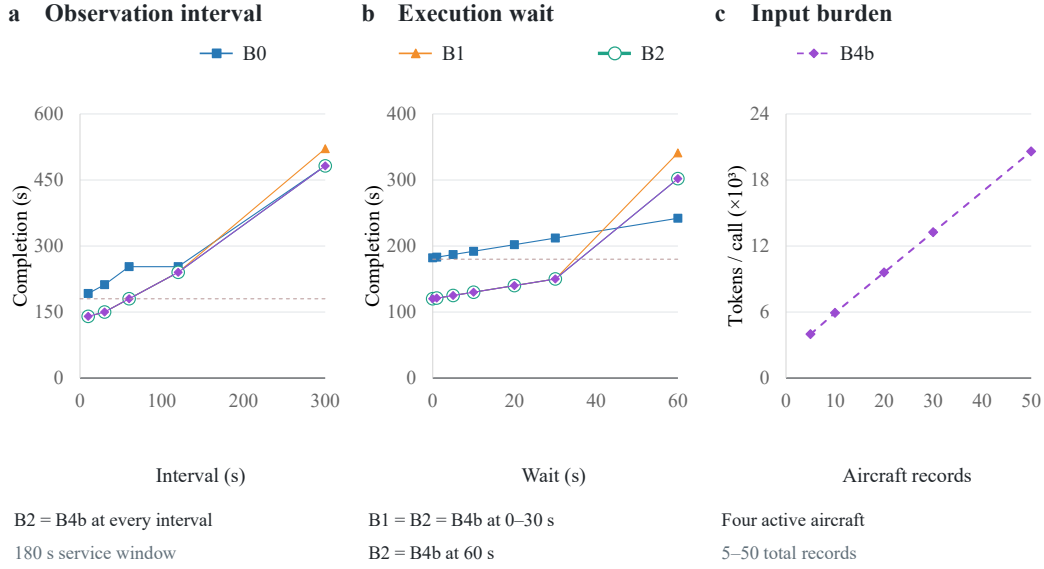


Figure 6. E4 timing and input burden at Site A. Panels (a,b) show mean completion duration for 20 runs per arm and policy. The dashed horizontal line marks the 180 s service window; straight segments join measured settings. B2/B4b coincide, as does B1 at 0–30 s in (b); nested markers and direct labels identify overlaps. Panel (c) shows prompt tokens per logged call for 5–50 records around four active aircraft.

The schedule delayed both task discovery and fault observation. The 300 s snapshot preceded the same-second task event; the task first appeared at 310/330/360/360/600 s across the five arms. The 371 s fault appeared at 380/390/420/480/600 s. OBS120 restored the affected path without timely completion. OBS300 had no affected primary air chain, so its conditional recovery was N/A. A timestamp alone did not describe event visibility.

Longer intervals also reduced B4b decisions from 62 to four per run. These counts cover the fixed full-horizon polling protocol, including calls after some services resolved. They do not estimate an adaptive policy that stops polling after completion.

6.6.2. *Execution waiting and command replacement*

Across injected waits of 0–30 s, B1/B2/B4b completion rose from 120 to 150 s and remained timely. At D60, B2/B4b completed in 302 s and B1 in 341 s; all B4b runs were late (Figure 6b; Supplementary Table S5).

The D60 sequence changed the surviving command (Supplementary Figure S2). An air proposal selected at 300 s waited until 360 s, when the fault and a newer ground decision superseded it. Ground fallback then waited until 420 s and completed at 602 s. This was command replacement, not checker rejection of a stale action. After correcting identical-command replacement, B0 instead executes its ground command at 360 s and completes at 542 s: 242 s from release. Its original 302 s duration included an unintended second 60 s wait. All 20 B0 D60 runs reproduce the correction; D00 and D30 controls retain 182 and 212 s, and neighboring fault times retain 242 s. Supplementary Figure S2 distinguishes this corrected ground-only sequence from B4b’s genuine air-to-ground replacement. These are effects of the specified event order and command semantics.

6.6.3. *Input burden with a fixed active fleet*

Increasing total aircraft records from five to fifty raised B4b prompt tokens per logged call from 3,982.1 to 20,593.5 (Figure 6c; Supplementary Table S6). The active fleet remained four aircraft; 640 matched-stage checks retained the same legal candidates. All 100 B4b runs completed on time, with no observed illegal or nonexistent resource selection.

The N05 completion mean was 121.5 s, compared with 120 s in N10–N50, alongside one retained transport error and another decision opportunity. E4 records four final transport-error calls and four structured retries overall. These remain included: the result concerns input burden for a fixed service problem, rather than coordination of fifty operational aircraft.

7. Discussion

7.1. *An executable service-management architecture*

We develop the Manager as a transport method whose decision object is a service intervention. Joint state becomes useful only when linked to a compatible resource, an executable action and a completion estimate. The candidate interface makes that link explicit and carries incumbent-service consequences into selection. Common checking and execution let researchers compare policy preferences without changing the underlying transport contract. This extends multimodal recovery and dynamic fleet management to a failure-prone ground–air service system. (Pillac et al. 2013; Jaber et al. 2025; Wollenstein-Betech et al. 2022)

The comparison shows why a replaceable policy layer is useful. Deterministic and LLM policies often obtain similar outcomes because the surviving alternatives strongly constrain what can be achieved. Where timely ground competes with faster preemptive air, policies can express different service preferences. B2 makes its valuation explicit through fixed weights; B4b sometimes exchanges incumbent preservation for additional emergency speed. The Manager exposes that exchange so an operator can evaluate it against service obligations.

7.2. *Useful alternatives and the remaining service margin*

Low-altitude capacity yields resilience when it supplies a useful alternative that can be executed in time. E1 demonstrates the bypass benefit and the cost of reallocating occupied aircraft. E2 separates successful path restoration from timely completion. E3 shows how surviving options protect priority-weighted services, while exhausted alternatives leave the same loss across policies. Together these findings connect network recovery to the service outcomes that motivate intervention. (Mattsson and Jenelius 2015; Cats and Jenelius 2018)

The decision-log analysis identifies contrasting operating conditions. Clear speed and service-preservation advantages tend to align choices; smaller speed gains with preemption expose preferences; negative predicted margins identify services that are difficult to rescue. Predictions remain approximate, as Site C’s few timely completions despite negative candidate margins illustrate. The interface must therefore expose the provenance of ETA and service consequences, and timing estimates must be interpreted together with realized trajectories.

Transfer sensitivity reinforces this mechanism. In the offline replacement analysis, Site B’s timely affected services retain only 19 s of margin. A 30 s handover assumption makes 80 previously timely services per coordinated policy late. Site A’s deterministic replacements retain 60 s, while a small number of LLM trajectories are closer to the deadline. This does not alter the observed path-restoration result, but it makes physical transfer timing central to deploying the architecture.

7.3. *Observation and execution are management decisions*

Observation scheduling determines when a mission or fault enters the decision state. Pending-command semantics determine whether an accepted intervention reaches execution before circumstances change. E4 demonstrates both mechanisms: sparse observation can exhaust the deadline, and waiting can allow a fault to replace an air command with ground service. The B0 duplicate-command correction removes an implementation-induced delay and leaves the substantive air-to-ground replacement mechanism intact. A practical Manager needs deadline-aware observation and idempotent handling of repeated commands, in addition to a capable selection policy.

Explicit candidate information also imposes computational input demand. B4a–B4b measures the support supplied by derived candidate facts, while E4C measures input burden around a fixed active fleet. These results guide interface design and policy selection together: the available facts should make consequential choices comparable without obscuring them in unnecessary records.

7.4. *Scope and next steps*

The evidence concerns fixed, site-adapted scenario panels and a four-aircraft core fleet. Road morphology varies together with facility placement, flight distance and fault timing. Ground service uses a corridor ETA timer, and the registry abstracts payload custody and transfer synchronization; offline added-time sensitivity does not re-optimize a closed loop. Faults and feasibility checks implement specified service-level rules. SWL uses configured flat penalties, and incumbent damage is a final-state count. The tested LLM is one recorded backend and prompt family, with wall-clock inference decoupled from simulated transport. Next steps are task-specific ground and payload-transfer models, closed-loop transfer-delay experiments, larger active fleets and controlled variation

of transport geometry. These extensions would test operational generalization of the mechanisms established here.

8. Conclusions

We propose a Ground–Low–Altitude Mobility Manager that connects joint transport observations to executable service reconfiguration. Its shared candidate interface makes feasibility, remaining time and incumbent-service consequences comparable across replaceable supervisory policies. The closed-loop experiments demonstrate selective air-support benefits, expose speed–service-preservation trade-offs and distinguish path restoration from timely service recovery.

Low-altitude capacity alone does not ensure resilience. Its value depends on selecting and executing useful cross-modal interventions before the remaining service margin is exhausted. Surviving alternatives define what can be rescued, service preferences determine which trade-offs are accepted, and observation–execution timing determines whether the selected intervention takes effect. The Manager provides a concrete architecture and comparative evidence for managing these dependencies.

Data and code availability

The accompanying local research package contains run artifacts, experiment configurations, prompt files, analysis scripts, revised figure data and an executable queue-correction overlay. Supplementary Section S6 identifies the files and reproduction commands. Original data and the previous manuscript version are retained separately. A public repository identifier has not yet been assigned.

References

- Alvarez Lopez, Pablo, Michael Behrisch, Laura Bieker-Walz, Jakob Erdmann, Y.-P. Flötteröd, Robert Hilbrich, Leonhard Lüken, Johannes Rummel, Peter Wagner, and Eva Wießner. 2018. “Microscopic Traffic Simulation using SUMO.” In *21st IEEE International Conference on Intelligent Transportation Systems*, 2575–2582.
- Cats, Oded, and Erik Jenelius. 2018. “Beyond a complete failure: the impact of partial capacity degradation on public transport network vulnerability.” *Transportmetrica B: Transport Dynamics* 6 (2): 77–96. <https://doi.org/10.1080/21680566.2016.1267596>.
- Federal Aviation Administration. 2023. *Urban Air Mobility Concept of Operations, Version 2.0*. Technical Report. Federal Aviation Administration. <https://www.faa.gov/sites/faa.gov/files/Urban-Air-Mobility-Concept-of-Operations-2.0.pdf>.
- Haklay, Mordechai, and Patrick Weber. 2008. “OpenStreetMap: User-Generated Street Maps.” *IEEE Pervasive Computing* 7 (4): 12–18. <https://discovery.ucl.ac.uk/id/eprint/13849/>.
- Hoekstra, Jacco M., and Joost Ellerbroek. 2016. “BlueSky ATC Simulator Project: An Open Data and Open Source Approach.” In *7th International Conference on Research in Air Transportation*, <https://resolver.tudelft.nl/uuid:d1131a90-f0ea-4489-a217-ad29987689a1>.
- Holm, Sture. 1979. “A Simple Sequentially Rejective Multiple Test Procedure.” *Scandinavian Journal of Statistics* 6 (2): 65–70. <https://www.jstor.org/stable/4615733>.
- Ichter, Brian, et al. 2023. “Do As I Can, Not As I Say: Grounding Language in Robotic

- Affordances.” In *Proceedings of the 6th Conference on Robot Learning*, Vol. 205, 287–318. <https://proceedings.mlr.press/v205/ichter23a.html>.
- Jaber, Sara, Mostafa Ameli, S. M. Hassan Mahdavi, and Neila Bhourri. 2025. “A methodological framework for Resilience as a Service (RaaS) in multimodal urban transportation networks.” *Sustainable Cities and Society* 124: 106276. <https://doi.org/10.1016/j.scs.2025.106276>.
- Jenelius, Erik, and Lars-Göran Mattsson. 2015. “Road network vulnerability analysis: Conceptualization, implementation and application.” *Computers, Environment and Urban Systems* 49: 136–147. <https://doi.org/10.1016/j.compenvurbsys.2014.02.003>.
- Lai, D. S. W., and J. M. Y. Leung. 2018. “Real-time rescheduling and disruption management for public transit.” *Transportmetrica B: Transport Dynamics* 6 (1): 17–33. <https://doi.org/10.1080/21680566.2017.1358678>.
- Lai, Shun, Zhen Xu, Wei Zhang, Hui Liu, and Huan Xiong. 2024. “LLMLight: Large Language Models as Traffic Signal Control Agents.” *arXiv preprint arXiv:2312.16044* Version 5, <https://arxiv.org/abs/2312.16044v5>.
- Liu, Nelson F., Kevin Lin, John Hewitt, Aditya Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. “Lost in the Middle: How Language Models Use Long Contexts.” *Transactions of the Association for Computational Linguistics* 12: 157–173. <https://aclanthology.org/2024.tacl-1.9/>.
- Mattsson, Lars-Göran, and Erik Jenelius. 2015. “Vulnerability and resilience of transport systems—A discussion of recent research.” *Transportation Research Part A: Policy and Practice* 81: 16–34. <https://doi.org/10.1016/j.tra.2015.06.002>.
- Meng, Shanshan, Dong Li, Jiyin Liu, and Yanru Chen. 2024. “The multi-visit drone-assisted routing problem with soft time windows and stochastic truck travel times.” *Transportation Research Part B: Methodological* 190: 103101. <https://doi.org/10.1016/j.trb.2024.103101>.
- Murray, Chase C., and Amanda G. Chu. 2015. “The flying sidekick traveling salesman problem: Optimization of drone-assisted parcel delivery.” *Transportation Research Part C: Emerging Technologies* 54: 86–109. <https://doi.org/10.1016/j.trc.2015.03.005>.
- Pillac, Victor, Michel Gendreau, Christophe Guéret, and Andrés L. Medaglia. 2013. “A review of dynamic vehicle routing problems.” *European Journal of Operational Research* 225 (1): 1–11. <https://doi.org/10.1016/j.ejor.2012.08.015>.
- Wabersich, Kim P., and Melanie N. Zeilinger. 2021. “A predictive safety filter for learning-based control of constrained nonlinear dynamical systems.” *Automatica* 129: 109597. <https://doi.org/10.1016/j.automatica.2021.109597>.
- Wollenstein-Betech, Salomón, Mauro Salazar, Arian Houshmand, Marco Pavone, Ioannis C. Paschalidis, and Christos G. Cassandras. 2022. “Routing and Rebalancing Intermodal Autonomous Mobility-on-Demand Systems in Mixed Traffic.” *IEEE Transactions on Intelligent Transportation Systems* 23 (8): 12263–12275. <https://doi.org/10.1109/TITS.2021.3112106>.
- Yap, Menno, Oded Cats, Johanna Törnquist Krasemann, Niels van Oort, and Serge Hoogenboom. 2022. “Quantification and control of disruption propagation in multi-level public transport networks.” *International Journal of Transportation Science and Technology* 11 (1): 83–106. <https://doi.org/10.1016/j.ijtst.2021.02.002>.